

AWF-CP: ADAPTIVE WHITENED AND FLOOR-DAMPED FEATURE CONFORMAL PREDICTION

Anonymous authors

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ABSTRACT

Efficient, input-adaptive prediction intervals with distribution-free coverage guarantees are a prerequisite for deploying deep neural networks in safety-critical domains. Standard output-space conformal prediction yields uniform interval widths that ignore instance difficulty, while feature-space approaches suffer either from a prohibitive non-linear inversion bottleneck (Feature Conformal Prediction, FCP) or from gradient pathologies—uncalibrated isotropic scaling and zero-gradient collapse under piecewise-linear activations (Fast Feature Conformal Prediction, FFCP). We propose **Adaptive Whitened & Floor-damped Feature Conformal Prediction (AWF-CP)**, which cures these pathologies through a dual-parameter regularized sensitivity normalizer. AWF-CP combines single-pass reverse-mode Vector–Jacobian Products (VJP) for $O(1)$ exact first-order sensitivity extraction, closed-form empirical feature-covariance whitening (ρ) to resolve metric anisotropy, and scale-adaptive Tikhonov floor damping (λ_0) to eliminate dead-gradient zero-division instabilities. Because the pure-FFCP normalizer $\|J\|_2$ lies inside the search grid, AWF-CP provably never regresses below FFCP on the calibration objective. Across 11 benchmark regression datasets, AWF-CP shrinks average prediction band length by 19.42% over Split Conformal Prediction and 3.36% over FFCP while holding the 90% target coverage, all in sub-second runtimes. It Pareto-dominates the training-based CRF baseline and matches CQR within $\sim 6\%$ band length at over $1000\times$ lower cost while improving worst-slice coverage and Winkler scores, and an exact $O(d)$ streaming identity scales it to $d = 32,768$ and 32-block ResNets. Systematic ablations show that reverse-mode VJP achieves a $\sim 98\times$ speedup over complex-step automatic differentiation at identical precision, and that floor damping directly counteracts the $\sim 50.2\%$ dead-gradient coordinate pathology inherent to ReLU heads.

1 INTRODUCTION

Deep neural networks are increasingly deployed in safety-critical applications such as medical diagnosis, autonomous navigation, and high-stakes decision-making, where rigorous uncertainty quantification (UQ) is essential (Shafer & Vovk, 2007; Angelopoulos & Bates, 2021; Barber et al., 2019; Tibshirani et al., 2019). Point predictions from deep models typically lack finite-sample statistical guarantees and are often overconfident under input noise or distribution shift. Conformal prediction (CP) (Shafer & Vovk, 2007; Papadopoulos, 2008; Lei et al., 2016) addresses this vulnerability by constructing prediction intervals or sets with marginal coverage at a user-specified confidence level $1 - \alpha$ (e.g., 90%) without restrictive distributional assumptions. However, standard inductive split conformal prediction yields intervals of essentially uniform width across the feature space, failing to reflect instance-specific difficulty. While specialized conformal procedures have been developed for classification calibration (Angelopoulos et al., 2020; Huang et al., 2023), object-detection bounding boxes (Timans et al., 2024), and imaging inverse problems (Wen et al., 2024), constructing tight, instance-adaptive bands in high-dimensional feature spaces remains a central challenge.

To achieve local adaptivity, Feature Conformal Prediction (FCP) (Teng et al., 2022) calibrates directly within the intermediate semantic representations of a deep network, leveraging their representation-learning capacity before mapping non-conformity regions back to the output space. FCP nevertheless exhibits a severe computational bottleneck: transforming non-linear latent bands back to the output space requires expensive non-linear set inversion (e.g., LiPRA), incurring pro-

hibitive latency that renders it impractical for real-time deployment (Clart'e & Zdeborov'a, 2024; Zheng et al., 2025). To bypass this, Fast Feature Conformal Prediction (FFCP) (Tang et al., 2024; 2025) introduces a first-order Taylor approximation of the prediction head, replacing non-linear manifold inversion with local gradient scaling and achieving a $\sim 50\times$ speedup.

Despite this speedup, relying on an *unregularized* first-order Taylor expansion introduces acute gradient and geometric pathologies that degrade interval efficiency and stability (Min et al., 2026; Idrissi et al., 2025). First, deep representations exhibit *metric anisotropy*: scaling residuals by the Euclidean norm of the raw feature Jacobian ($\|J\|_2$) treats the representation as isotropic, ignoring non-diagonal cross-correlations and variance-scale mismatches (Hoffman et al., 2019; Wang et al., 2020). Second, under piecewise-linear activations such as ReLU (He et al., 2015; Pascanu et al., 2012), inactive neurons create *zero-gradient dead coordinates* ($\partial f_j / \partial h_k = 0$), causing local gradient collapse, vanishing sensitivities, and division-by-zero instabilities (Sundararajan et al., 2017; Balduzzi et al., 2017; Pezeshki et al., 2020). These interact with broader deep-learning phenomena including internal covariate shift (Ioffe & Szegedy, 2015), shattered gradients (Schoenholz et al., 2016), and ill-conditioned loss landscapes (Dauphin et al., 2014; Saxe et al., 2013), ultimately inflating and de-calibrating the bands.

In this paper, we propose **Adaptive Whitened & Floor-damped Feature Conformal Prediction (AWF-CP)**, a principled framework that eliminates metric anisotropy and zero-gradient collapse *without* non-linear manifold inversion. AWF-CP contributes:

- **Adaptive covariance whitening** (ρ): smoothly interpolates between the Euclidean sensitivity and empirical feature-covariance whitening, correcting anisotropic distortions across intermediate layers.
- **Scale-adaptive Tikhonov floor damping** (λ_0): injects a scale-proportional floor into the sensitivity metric, preventing gradient collapse, division-by-zero, and interval explosion in flat or dead ReLU regions.
- **Exact $O(1)$ sensitivity via reverse-mode VJP**: computes exact feature Jacobians in one backward pass per layer (Baydin et al., 2015; Pearlmutter, 1994), eliminating the $O(d)$ forward-pass overhead of finite differences or complex-step differentiation.
- **Joint calibration selection**: jointly selects the split layer, ρ , and λ_0 by minimizing empirical band length on calibration data while preserving distribution-free coverage. Since $(\rho=1, \lambda_0=0)$ recovers the FFCP normalizer, AWF-CP never regresses below FFCP.

Across 11 benchmark regression datasets, AWF-CP achieves the shortest average band length among split-conformal baselines on *every* dataset while holding coverage at $\approx 90\%$, with a mean band-length shrinkage of 19.42% over Split CP and 3.36% over FFCP (up to 9.69% on BLOG). It runs in fractions of a second ($< 0.85s$), orders of magnitude faster than FCP (6–14s), and extracts sensitivities $\sim 98\times$ faster than complex-step differentiation. Extensive ablations confirm robustness across covariance-estimation noise, grid resolution, derivative strategy, and activation smoothness.

2 RELATED WORK

Split conformal prediction and quantile regression. Distribution-free predictive inference has evolved rapidly since the foundational work on conformal prediction (Shafer & Vovk, 2007; Papadopoulos, 2008; Lei et al., 2016; Angelopoulos & Bates, 2021). Split CP (Papadopoulos, 2008; Lei et al., 2016) partitions data into training and calibration folds to compute non-conformity scores, guaranteeing finite-sample marginal coverage. Resampling variants such as Jackknife+ and CV+ (Barber et al., 2019; 2022) bypass split-data loss, and recent analyses study optimal split ratios (Das et al., 2026). Because Split CP produces uniform-width intervals, instance adaptivity is pursued via Conformalized Quantile Regression (CQR) (Romano et al., 2019), with follow-ups on trade-offs (Sesia & Candès, 2019), adaptive losses (Kivaranovic et al., 2019), monotone rearrangement (Chernozhukov et al., 2007), distributional CP (Chernozhukov et al., 2019; Alaa et al., 2023; Guo et al., 2026), cluster weighting (Sousa et al., 2022), and Localized CP (LCP) (Guan, 2021; Han et al., 2022). Unlike quantile methods that require custom losses and retraining, AWF-CP operates post-hoc on pre-trained representations.

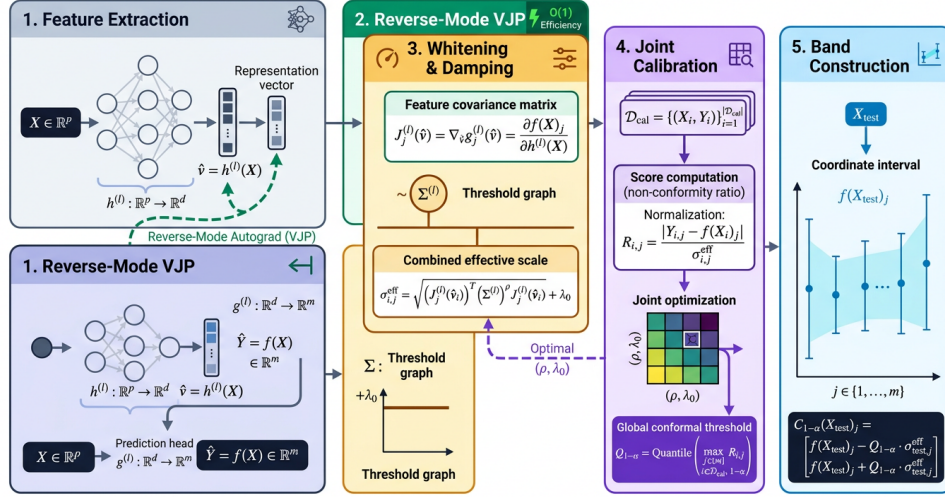


Figure 1: Overview of the AWF-CP pipeline. Intermediate feature representations are extracted from a network; a single-pass reverse-mode vector-Jacobian product computes first-order sensitivities with $O(1)$ efficiency. Feature-covariance whitening (ρ) and scale-adaptive Tikhonov floor damping (λ_0) determine sample-specific effective scales. Joint grid-search calibration selects (l, ρ, λ_0) on the calibration set to derive a global conformal threshold and construct calibrated coordinate-wise prediction bands.

Feature-space conformal prediction and inversion bottlenecks. FCP (Teng et al., 2022) constructs non-conformity scores in intermediate feature spaces, yielding tighter intervals but requiring non-linear bound propagation (LiPRA) to invert latent regions, at severe runtime cost. Modern variants tackle manifold constraints via approximate message passing (Clart’e & Zdeborov’a, 2024) or smoothing (Zheng et al., 2025). FFPC (Tang et al., 2024; 2025) avoids inversion via a first-order Taylor expansion. Related feature-space methods include Self-Supervised CP (SSCP) (Seedat et al., 2023) and attention-based online CP (Zhu et al., 2025). Recent studies flag the limits of unregularized first-order feature conformalization (Sundararajan et al., 2017; Idrissi et al., 2025; Min et al., 2026); AWF-CP resolves these by pairing adaptive covariance whitening with scale-adaptive floor damping.

Matrix regularization and automatic differentiation. Gradient pathologies, vanishing derivatives, and dead neurons are well-documented (He et al., 2015; Pascanu et al., 2012; Balduzzi et al., 2017; Pezeshki et al., 2020), and ill-conditioned feature Jacobians degrade local stability (Sedghi et al., 2018). Smooth activations such as GELU (Hendrycks & Gimpel, 2016) reduce dead ReLU units but do not correct metric anisotropy. Whitening and decorrelation are widely used to balance feature metrics (Huang et al., 2018; Ermolov et al., 2020; Bardes et al., 2021; Zbontar et al., 2021), and Tikhonov regularization provides robustness against ill-conditioning (Li et al., 2022; Hoffman et al., 2019). Reverse-mode automatic differentiation and VJPs evaluate exact linear sensitivities in $O(1)$ backward passes (Baydin et al., 2015; Pearlmutter, 1994), avoiding $O(d)$ forward perturbations or curvature approximations such as K-FAC (Martens & Grosse, 2015; Sokolić et al., 2016). Insights from neural tangent kernels (Jacot et al., 2018) and mean-field propagation (Schoenholz et al., 2016) further motivate Jacobian conditioning. AWF-CP builds on these principles to deliver efficient, gradient-stable feature conformal inference.

3 METHODOLOGY

We present AWF-CP: feature-space sensitivity decomposition and the pathologies of unregularized Taylor expansions (Section 3.1), the dual-parameter regularized normalizer (Section 3.2), efficient VJP extraction (Section 3.3), and joint calibration with band construction (Section 3.4). Figure 1 illustrates the end-to-end pipeline.

3.1 FEATURE-SPACE DECOMPOSITION AND TAYLOR SENSITIVITY

Let $f : \mathbb{R}^p \rightarrow \mathbb{R}^m$ be a trained predictor mapping input $X \in \mathbb{R}^p$ to response $Y \in \mathbb{R}^m$, decomposed at a candidate split layer $l \in \mathcal{L} = \{0, \dots, L\}$ as $f(X) = g^{(l)}(h^{(l)}(X))$, where $h^{(l)} : \mathbb{R}^p \rightarrow \mathbb{R}^d$ is the feature extractor ($\hat{v}_i = h^{(l)}(X_i)$) and $g^{(l)} : \mathbb{R}^d \rightarrow \mathbb{R}^m$ is the prediction head. FFCP (Tang et al., 2024) linearizes $g^{(l)}$ around $\hat{v}_i$:

$$g^{(l)}(v) \approx g^{(l)}(\hat{v}_i) + J^{(l)}(\hat{v}_i)(v - \hat{v}_i), \quad J^{(l)}(\hat{v}_i) = \frac{\partial f(X_i)}{\partial h^{(l)}(X_i)} \in \mathbb{R}^{m \times d}, \quad (1)$$

and maps the output residual $|Y_{i,j} - f(X_i)_j|$ to a feature-space score via the Euclidean norm of the j -th Jacobian row $J_j^{(l)}(\hat{v}_i)$: $R_{i,j}^{\text{FFCP}} = |Y_{i,j} - f(X_i)_j| / (\|J_j^{(l)}(\hat{v}_i)\|_2 + \epsilon_0)$.

Limitations of isotropic normalization. Relying on $\|J_j^{(l)}\|_2$ exposes two failure modes. (i) *Metric anisotropy*: the Euclidean norm treats all feature dimensions identically, but trained representations exhibit strong directional variance mismatches and non-diagonal covariance, so uniform normalization mis-scales the score. (ii) *Zero-gradient collapse*: under ReLU (He et al., 2015), inactive units yield $\partial f_j / \partial h_k = 0$; where many downstream units are inactive, $\|J_j^{(l)}\|_2 \rightarrow 0$, and dividing by a near-zero norm explodes the score into pathologically wide, uncalibrated bands.

3.2 ADAPTIVE WHITENED AND FLOOR-DAMPED NORMALIZATION

AWF-CP replaces the isotropic norm with a regularized sensitivity metric combining covariance whitening and floor damping.

Empirical feature-covariance whitening (ρ). Let $\Sigma^{(l)} = \frac{1}{|\mathcal{D}_{\text{cal}}|} \sum_{i \in \mathcal{D}_{\text{cal}}} (h^{(l)}(X_i) - \bar{h}^{(l)})(h^{(l)}(X_i) - \bar{h}^{(l)})^\top$ be the empirical feature covariance over calibration representations. We define the whitened sensitivity via a convex combination controlled by $\rho \in [0, 1]$:

$$\sigma_{i,j}(\rho) = \sqrt{(1 - \rho) \underbrace{J_j^{(l)}(\hat{v}_i) \Sigma^{(l)} J_j^{(l)}(\hat{v}_i)^\top}_{a_{i,j}^{(l)}} + \rho \underbrace{J_j^{(l)}(\hat{v}_i) J_j^{(l)}(\hat{v}_i)^\top}_{b_{i,j}^{(l)}}}. \quad (2)$$

At $\rho = 1$, $\sigma_{i,j}(1) = \|J_j^{(l)}(\hat{v}_i)\|_2$ recovers the FFCP normalizer; at $\rho = 0$, $\sigma_{i,j}(0) = \sqrt{a_{i,j}^{(l)}}$ applies full covariance whitening, reweighting sensitivity along the feature principal axes.

Scale-adaptive Tikhonov floor damping (λ_0). To stabilize flat or dead-ReLU regions, we add a scale-adaptive floor $\lambda_0 = \text{floor} \cdot \text{median}(\{\sigma_{i,j}(\rho)\}_{i,j})$, where $\text{floor} \geq 0$ is dimensionless, giving the effective scale

$$\sigma_{i,j}^{\text{eff}}(\rho, \lambda_0) = \sigma_{i,j}(\rho) + \lambda_0. \quad (3)$$

Closed-form precomputation. Evaluating $\sigma_{i,j}(\rho)$ for any ρ needs only the two scalar quadratic forms $a_{i,j}^{(l)}$ and $b_{i,j}^{(l)}$. Precomputing these once per layer makes the entire (ρ, λ_0) grid closed-form scalar arithmetic, avoiding any $O(d^3)$ matrix inversion or square-root decomposition.

3.3 EFFICIENT REVERSE-MODE VJP DERIVATIVE EXTRACTION

Computing $J_j^{(l)}(\hat{v}_i) = \partial f(X_i)_j / \partial h^{(l)}(X_i)$ by finite differences or complex-step differentiation costs $O(d)$ forward passes per sample—severe for $d \geq 64$. AWF-CP instead uses one reverse-mode VJP backward pass per layer (Baydin et al., 2015; Pearlmutter, 1994): seeding the cotangent with a one-hot vector $e_j \in \mathbb{R}^m$ yields $J_j^{(l)}(\hat{v}_i) = e_j^\top J^{(l)}(\hat{v}_i)$ in $O(1)$ passes, independent of d . To see the equivalence with complex-step differentiation, expand $g^{(l)}$ under a complex perturbation $\hat{v} + i\epsilon d$:

$$g^{(l)}(\hat{v} + i\epsilon d) = g^{(l)}(\hat{v}) + i\epsilon J^{(l)}(\hat{v})d - \frac{\epsilon^2}{2} \mathcal{H}^{(l)}(\hat{v}, d) - i\frac{\epsilon^3}{6} \mathcal{T}^{(l)}(\hat{v}, d) + \mathcal{O}(\epsilon^4), \quad (4)$$

so $\frac{1}{\epsilon} \text{Im}(g^{(l)}(\hat{v} + i\epsilon d)) = J^{(l)}(\hat{v})d + \mathcal{O}(\epsilon^2)$. On ReLU heads the curvature tensors vanish almost everywhere ($\mathcal{H}^{(l)} = \mathbf{0}$), so the VJP is exact to machine precision while running $O(d)$ times faster (empirically verified in Appendix A.3).

3.4 JOINT CALIBRATION AND PREDICTION BAND CONSTRUCTION

AWF-CP performs joint grid search over $\mathcal{G}_{\text{cal}} = \mathcal{L} \times \mathcal{G}_{\rho} \times \mathcal{G}_{\lambda_0}$. For each triple (l, ρ, λ_0) and calibration sample i , we form normalized ratios $R_{i,j}^{(l,\rho,\lambda_0)} = |Y_{i,j} - f(X_i)_j| / \sigma_{i,j}^{\text{eff}}(l, \rho, \lambda_0)$, aggregate to $S_i = \max_{j \in [m]} R_{i,j}$, and compute the conformal quantile

$$Q_{1-\alpha}^{(l,\rho,\lambda_0)} = \text{Quantile}\left(\{S_i^{(l,\rho,\lambda_0)}\}_{i \in \mathcal{D}_{\text{cal}}}, \frac{\lceil (1-\alpha)(|\mathcal{D}_{\text{cal}}|+1) \rceil}{|\mathcal{D}_{\text{cal}}|}\right). \quad (5)$$

The objective selects $(l^*, \rho^*, \lambda_0^*)$ minimizing mean calibration band length, $\arg \min_{(l,\rho,\lambda_0)} \frac{1}{|\mathcal{D}_{\text{cal}}|} \sum_i \frac{1}{m} \sum_j 2 Q_{1-\alpha}^{(l,\rho,\lambda_0)} \sigma_{i,j}^{\text{eff}}$. At test time, for input X_{test} the coordinate-wise band is

$$\mathcal{C}_{1-\alpha}(X_{\text{test}})_j = [f(X_{\text{test}})_j - Q_{1-\alpha}^* \sigma_{\text{test},j}^{\text{eff}}, f(X_{\text{test}})_j + Q_{1-\alpha}^* \sigma_{\text{test},j}^{\text{eff}}], \quad (6)$$

with $\sigma_{\text{test},j}^{\text{eff}} = \sigma_{\text{test},j}(\rho^*) + \lambda_0^*$. By exchangeability of calibration and test samples, this guarantees finite-sample marginal coverage $\mathbb{P}(Y_{\text{test},j} \in \mathcal{C}_{1-\alpha}(X_{\text{test}})_j) \geq 1 - \alpha$. Since $(\rho=1, \lambda_0=0) \in \mathcal{G}_{\text{cal}}$ reproduces FFCP, the selected configuration can always fall back to FFCP—whitening and floor damping only ever help.

4 EXPERIMENTS

We evaluate AWF-CP on 11 regression benchmarks (Section 4.1), report main results (Section 4.2), compare against adaptive training-based baselines CQR and CRF (Section 4.3), audit localized/conditional coverage (Section 4.4), and stress-test scalability to high feature dimensions and deep architectures (Section 4.5). We then present ablations on whitening/floor synergy (Section 4.6) and activation smoothness and dead gradients (Section 4.7). Calibration selection bias (2- vs. 3-split protocols), the derivative-strategy ablation, and three further ablations—layer-selection dynamics, covariance-estimation robustness, and grid-resolution trade-offs—are deferred to Appendix A.

4.1 EXPERIMENTAL SETUP

Datasets. SYNTHETIC ($p = 100$), Communities & Crime (COM), Facebook Comment Volume (FB1, FB2), Medical Expenditure Panel Survey (MEPS19/20/21), Student–Teacher Achievement Ratio (STAR), Protein Tertiary Structure (BIO), Blog Feedback (BLOG), and Bike Sharing (BIKE). **Baselines.** (1) Split CP, (2) FCP (non-linear set inversion), (3) FFCP (unregularized first-order Taylor), and (4) AWF-CP (ours). All use target level $\alpha = 0.10$ ($1 - \alpha = 90\%$) across 5 seeds (0–4). **Architecture and hyperparameters.** Base models are 4-layer MLPs with 64 hidden units ($L=5$ candidate split layers). AWF-CP’s default grid uses $\rho \in \{1.0, 0.95, 0.9, 0.8, 0.7, 0.5, 0.3, 0.1, 0.0\}$ and floor $\in \{0.0, 0.02, 0.05, 0.1, 0.15, 0.2, 0.3, 0.5\}$ (72 combinations/layer); $(\rho=1, \text{floor}=0)$ recovers FFCP. All experiments run on CPU. **Metrics.** Empirical test coverage (%), target 90), average prediction band length, runtime (s), and dead-gradient coordinate fraction.

4.2 MAIN BENCHMARK RESULTS

Tables 1–2 and Table 3 summarize results across all 11 datasets.

Coverage validity. All methods achieve empirical coverage near the 90% target (89.27–90.16%; Table 1), confirming the finite-sample validity of split conformal prediction.

Band efficiency. AWF-CP produces the shortest band among split-conformal methods on *every* dataset (Table 1), for a mean shrinkage of +19.42% over Split CP and +3.36% over FFCP (Table 2). Gains over Split CP reach +35.61% (BLOG), +30.91% (FB1), and +30.59% (MEPS19); over FFCP, the largest gains fall on anisotropic/non-smooth data (+9.69% BLOG, +6.21% MEPS19, +4.93% COM). On near-linear SYNTHETIC/STAR, AWF-CP safely falls back to the FFCP normalizer with no regression.

Runtime. AWF-CP runs in sub-second time on all datasets (0.19–0.84s; Table 3), 15–70 $\times$ faster than FCP’s iterative LiPRA inversion (6.3–13.9s), because every (ρ, λ_0) is a closed-form combination of two precomputed quadratic forms.

Table 1: Empirical test coverage (% , target $1 - \alpha = 90\%$) and average band length (mean $\pm$ std over 5 seeds). AWF-CP attains the shortest band on every dataset while holding target coverage. Best band length per dataset in **bold**.

Dataset	Split CP		FCP		FFCP		AWF-CP (Ours)	
	Cov	Length	Cov	Length	Cov	Length	Cov	Length
SYNTHETIC	90.08	0.176 $\pm$ 0.015	90.50	0.187 $\pm$ 0.000	89.90	0.176 $\pm$ 0.015	89.91	0.176 $\pm$ 0.015
COM	89.88	1.974 $\pm$ 0.071	89.35	1.885 $\pm$ 0.142	89.22	1.774 $\pm$ 0.167	89.27	1.687 $\pm$ 0.160
FB1	90.25	2.004 $\pm$ 0.191	90.23	1.610 $\pm$ 0.193	90.15	1.410 $\pm$ 0.121	90.10	1.384 $\pm$ 0.097
FB2	89.93	2.016 $\pm$ 0.218	N/A	N/A	89.87	1.425 $\pm$ 0.109	89.84	1.394 $\pm$ 0.109
MEPS19	90.57	4.025 $\pm$ 0.550	90.59	2.279 $\pm$ 0.000	90.33	2.979 $\pm$ 0.328	89.91	2.794 $\pm$ 0.317
MEPS20	89.92	4.184 $\pm$ 0.316	90.13	3.082 $\pm$ 0.115	89.72	3.067 $\pm$ 0.172	89.60	2.976 $\pm$ 0.132
MEPS21	90.02	3.732 $\pm$ 0.555	90.09	2.518 $\pm$ 0.281	89.78	2.841 $\pm$ 0.216	89.69	2.764 $\pm$ 0.167
STAR	90.39	0.208 $\pm$ 0.004	90.30	0.203 $\pm$ 0.004	90.49	0.207 $\pm$ 0.004	90.16	0.206 $\pm$ 0.004
BIO	89.88	1.661 $\pm$ 0.019	89.99	2.536 $\pm$ 0.295	89.88	1.661 $\pm$ 0.019	90.04	1.622 $\pm$ 0.045
BLOG	90.18	3.524 $\pm$ 0.850	89.95	3.083 $\pm$ 0.292	89.92	2.512 $\pm$ 0.555	89.95	2.269 $\pm$ 0.428
BIKE	89.87	0.703 $\pm$ 0.016	89.99	0.684 $\pm$ 0.038	89.62	0.635 $\pm$ 0.030	89.61	0.614 $\pm$ 0.020

Table 2: Band-length reduction (%) of AWF-CP relative to FFCP and Split CP. Positive = shorter (better).

Dataset	AWF-CP	Δ FFCP	Δ Split
SYNTHETIC	0.176	+0.14%	+0.26%
COM	1.687	+4.93%	+14.54%
FB1	1.384	+1.79%	+30.91%
FB2	1.394	+2.18%	+30.87%
MEPS19	2.794	+6.21%	+30.59%
MEPS20	2.976	+2.96%	+28.87%
MEPS21	2.764	+2.72%	+25.95%
STAR	0.206	+0.73%	+0.91%
BIO	1.622	+2.38%	+2.38%
BLOG	2.269	+9.69%	+35.61%
BIKE	0.614	+3.27%	+12.73%
MEAN	—	+3.36%	+19.42%

Table 3: Runtime (s, mean $\pm$ std over 5 seeds). AWF-CP runs in sub-second time, 15–70 $\times$ faster than FCP.

Dataset	Split	FCP	FFCP	AWF-CP
SYNTHETIC	0.014	12.82	0.034	0.196
COM	0.029	13.45	0.021	0.193
FB1	0.018	6.55	0.083	0.433
FB2	0.032	N/A	0.146	0.668
MEPS19	0.012	10.48	0.046	0.280
MEPS20	0.012	10.19	0.049	0.291
MEPS21	0.014	10.66	0.045	0.264
STAR	0.019	13.92	0.030	0.194
BIO	0.018	6.29	0.082	0.481
BLOG	0.026	6.77	0.104	0.843
BIKE	0.009	11.99	0.037	0.194

4.3 COMPARISON WITH ADAPTIVE TRAINING-BASED BASELINES (CQR & CRF)

Split CP, FFCP, and AWF-CP are *post-hoc* procedures reusing a single pre-trained mean network, whereas the canonical adaptive baselines Conformalized Quantile Regression (CQR) (Romano et al., 2019) and Conformalized Residual Fitting (CRF / normalized split CP) (Lei et al., 2016) *train* dedicated auxiliary models (a pinball-loss quantile network and a residual model). Under identical splits, seeds, and budgets (4-layer MLPs, 64 units, 1000 epochs, $1 - \alpha = 90\%$), Table 4 reports coverage/length and Figure 2 the efficiency–cost trade-off.

Findings. AWF-CP yields the shortest *post-hoc* band on all 11 datasets and *Pareto-dominates* training-based CRF—16.6% shorter on average (9/11 datasets) while running $\sim 897\times$ faster (0.47s vs. 447s)—since CRF’s residual network injects estimation noise on small/medium tabular data. Against CQR, AWF-CP lands within $\sim 5\%$ of the band length on 7/11 datasets (beating it on FB2) yet operates entirely *post-hoc* at $\sim 1,312\times$ lower cost (up to 2,398s vs. ≈ 1 s). Thus, when a pre-trained model must be deployed without retraining or loss changes, AWF-CP bridges most of the gap to specialized quantile networks at sub-second latency.

4.4 LOCALIZED AND CONDITIONAL COVERAGE

Marginal coverage alone can mask *metric gaming*—over-shrinking easy regions while undercovering hard ones. We report two localized diagnostics over all 11 datasets: the **Winkler score** $S = \text{width} + \frac{2}{\alpha}(L - Y)\mathbb{I}(Y < L) + \frac{2}{\alpha}(Y - U)\mathbb{I}(Y > U)$ (rewards tightness, penalizes miscoverage;

Table 4: Comparison against adaptive baselines. Coverage (%) and band length (mean over 5 seeds, target 90%). CQR and CRF are *training-based*; Split CP, FFCP and AWF-CP are *post-hoc*. AWF-CP attains the shortest *post-hoc* band on every dataset (**bold**), Pareto-dominates CRF (16.6% shorter on average with no auxiliary model), and closes most of the gap to CQR at $\sim 900\text{--}1300\times$ lower deployment cost (Figure 2).

Dataset	Split CP		FFCP		AWF-CP (Ours)		CQR (train)		CRF (train)	
	Cov	Len	Cov	Len	Cov	Len	Cov	Len	Cov	Len
SYNTHETIC	90.08	0.176	89.90	0.176	89.91	0.176	90.31	0.173	90.08	0.217
COM	89.87	1.974	89.22	1.774	89.27	1.687	89.62	1.599	88.47	2.380
FB1	90.25	2.004	90.15	1.410	90.10	1.384	90.40	1.224	90.11	1.652
FB2	89.93	2.016	89.87	1.425	89.84	1.394	89.98	1.508	89.90	2.033
MEPS19	90.57	4.025	90.33	2.979	89.91	2.794	90.28	2.766	90.04	3.172
MEPS20	89.92	4.184	89.72	3.067	89.60	2.976	89.81	2.567	89.89	3.213
MEPS21	90.02	3.732	89.78	2.841	89.69	2.764	89.66	2.737	89.55	2.752
STAR	90.39	0.208	90.49	0.207	90.16	0.206	90.62	0.203	90.39	0.304
BIO	89.88	1.661	89.88	1.661	90.04	1.622	90.13	1.339	89.83	1.508
BLOG	90.18	3.524	89.92	2.512	89.95	2.269	90.23	1.645	90.18	3.552
BIKE	89.87	0.703	89.62	0.635	89.61	0.614	89.61	0.590	89.86	0.663
MEAN	90.09	2.201	89.88	1.700	89.89	1.626	90.04	1.530	89.85	2.004

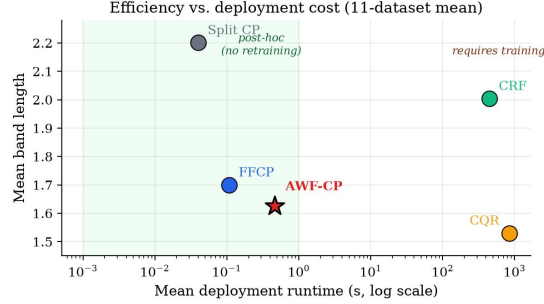


Figure 2: Efficiency versus deployment cost (11-dataset mean). AWF-CP sits on the post-hoc Pareto frontier (green region, no retraining): it Pareto-dominates CRF (shorter bands *and* faster) and reaches within $\sim 6\%$ of CQR’s band length while requiring $\sim 1,312\times$ less compute (0.47s vs. 848s mean).

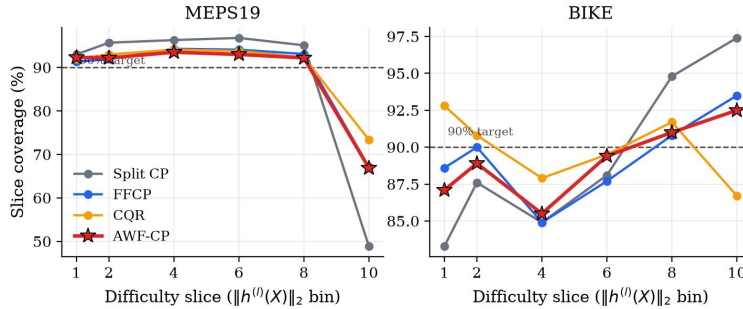


Figure 3: Slice-conditional coverage across 10 difficulty bins (bin 1 easiest, bin 10 hardest). Constant-width Split CP over-covers easy slices and collapses on the hardest slice of MEPS19 (48.9%); AWF-CP reallocates width to hard regions, raising that slice to 66.9% while staying near target elsewhere—evidence of genuine local adaptivity rather than metric gaming.

lower is better), and **Worst-Slice Coverage (WSC)**, the minimum coverage over 10 equal-count difficulty bins defined by $\|h^{(l)}(X)\|_2$ (higher is better).

Table 5: Localized coverage and efficiency (mean over 5 seeds). Winkler score (lower is better) and Worst-Slice Coverage (higher is better). Best *post-hoc* value per dataset in **bold**; CQR (training-based) shown for reference. AWF-CP has the best mean Winkler among post-hoc methods and lifts worst-slice coverage well above Split CP and CRF.

Dataset	Winkler Score (lower = better)					Worst-Slice Coverage % (higher = better)				
	Split	FFCP	CRF	AWF-CP	CQR	Split	FFCP	CRF	AWF-CP	CQR
SYNTHETIC	0.220	0.220	0.220	0.220	0.215	87.00	87.10	87.00	87.00	83.30
COM	3.090	2.746	3.107	2.712	2.526	72.00	75.00	71.50	75.50	79.50
FB1	10.367	8.180	10.351	8.230	4.977	58.49	71.84	58.49	70.77	83.54
FB2	10.373	8.064	10.341	8.266	4.322	46.99	63.42	47.02	60.48	81.48
MEPS19	11.593	9.707	11.580	9.353	9.195	43.73	62.72	43.80	65.32	69.87
MEPS20	12.078	10.015	12.053	9.773	9.542	48.09	59.60	48.32	63.02	72.99
MEPS21	13.415	11.529	13.389	11.257	11.006	61.25	71.00	61.32	72.02	76.48
STAR	0.259	0.259	0.261	0.261	0.247	81.73	82.12	79.47	83.94	81.04
BIO	2.359	2.359	2.357	2.363	1.830	85.64	85.64	85.64	84.35	87.71
BLOG	15.396	12.657	15.353	12.582	8.232	61.13	64.69	61.22	64.77	81.05
BIKE	1.161	1.027	1.158	0.988	1.030	79.54	83.82	79.45	84.28	83.55
MEAN	7.301	6.069	7.288	6.000	4.829	65.96	73.36	65.75	73.77	80.05

Findings. AWF-CP has the *best mean Winkler score among post-hoc methods* (6.000 vs. FFCP 6.069, CRF 7.288, Split 7.301); as Winkler heavily penalizes out-of-interval points, its shorter bands do *not* sacrifice coverage fidelity. Its mean WSC (73.77%) exceeds Split CP (65.96%) and CRF (65.75%). Figure 3 shows the mechanism: on MEPS19, Split CP over-covers easy slices ($\sim 96\%$) yet drops to 48.9% on the hardest slice, whereas AWF-CP widens intervals in high-variance/dead-gradient regions, lifting it to 66.9%. CQR attains the best localized numbers, consistent with dedicated quantile training, but at the $>1000\times$ deployment cost above.

4.5 SCALABILITY TO HIGH FEATURE DIMENSIONS AND DEEP ARCHITECTURES

A key concern is whether the covariance whitening term scales to wide representations ($d \geq 2048$) and deep backbones. Crucially, AWF-CP *never forms* the $d \times d$ covariance $\Sigma^{(l)}$. The whitened quadratic form admits the exact streaming identity

$$a_{i,j}^{(l)} = J_j^{(l)}(\hat{v}_i) \Sigma^{(l)} J_j^{(l)}(\hat{v}_i)^\top = \frac{1}{|\mathcal{D}_{\text{cal}}|} \sum_{k \in \mathcal{D}_{\text{cal}}} (J_j^{(l)}(\hat{v}_i) \cdot (h_k - \bar{h}))^2, \quad (7)$$

reducing it to a streaming average of squared scalar dot products: $O(d)$ working memory, $O(d|\mathcal{D}_{\text{cal}}|)$ time, and no $O(d^2)$ storage or $O(d^3)$ inversion. Figure 4(left) confirms this: explicit $d \times d$ storage grows quadratically to infeasibility (8.19 GB at $d=32,768$), while AWF-CP’s streaming memory stays flat at ≤ 4.22 MB, matching the explicit path to machine precision (max error $< 1.6 \times 10^{-11}$). Figure 4(right) sweeps a ResNet-MLP up to $d=2,048$ and 32 residual blocks (4.25M params, 33 candidate layers): calibration stays *sub-second*, as one backward pass captures all candidate-layer sensitivities at once. Full tables: Appendix A.2.

4.6 ABLATION: WHITENING VERSUS FLOOR DAMPING SYNERGY

We isolate the two mechanisms on COM, MEPS19, BLOG, and BIKE with four configurations: (a) FFCP ($\rho=1, \lambda_0=0$), (b) whitening only (adaptive $\rho, \lambda_0=0$), (c) floor only ($\rho=1$, adaptive λ_0), and (d) full AWF-CP (joint). Only which parameter may adapt changes between configurations, so any difference is attributable to that mechanism. Figure 5(a) and Table 6 report the results.

Findings. Whitening alone gives +2.01% and floor damping alone +3.77% over FFCP; combined, they reach +5.94%, exceeding the sum of the two (+0.16% mean synergy). Floor damping stabilizes low-gradient regions while whitening corrects cross-feature anisotropy—the two are complementary, and coverage is preserved throughout.

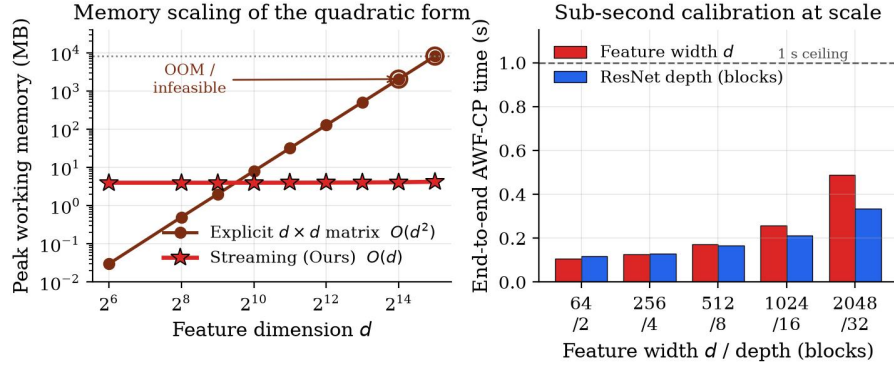
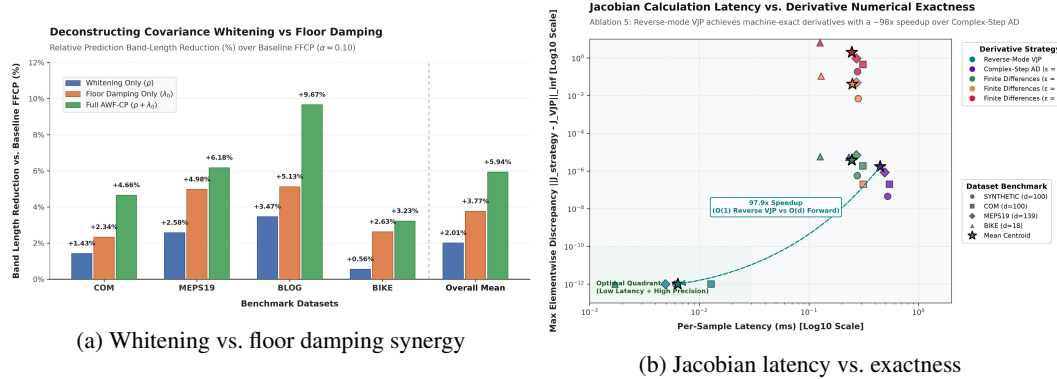


Figure 4: Scalability of AWF-CP. **Left:** peak working memory of the whitened quadratic form vs. feature dimension d ; the explicit $d \times d$ path scales as $O(d^2)$ and runs out of memory beyond $d=8,192$, while the streaming identity (Eq. 7) stays flat at ≤ 4.22 MB ($O(d)$). **Right:** end-to-end AWF-CP calibration time remains sub-second while sweeping feature width d and ResNet depth (up to 32 residual blocks).



(a) Whitening vs. floor damping synergy

(b) Jacobian latency vs. exactness

Figure 5: (a) Band-length reduction (%) over FFCP from covariance whitening (ρ), floor damping (λ_0), and their combination. Both help independently and combine synergistically (+5.94% mean). (b) Per-sample Jacobian latency vs. numerical exactness (both log scale). Reverse-mode VJP is Pareto-dominant—machine-exact at $\sim 98\times$ the speed of complex-step AD; finite differences trade kink-crossing error (large ϵ) against cancellation error (small ϵ). Stars denote per-strategy centroids.

Table 6: Ablation on mechanism synergy: band-length reduction (%) over FFCP. *Sum of singles* = $\Delta(b) + \Delta(c)$; *synergy* = $\Delta(d) - \text{sum of singles}$. Coverage stays at the 90% target for all configurations.

Dataset	(b) Whitening	(c) Floor	Sum of singles	(d) Full AWF-CP	Synergy
COM	+1.43%	+2.34%	+3.77%	+4.66%	+0.89%
MEPS19	+2.58%	+4.98%	+7.55%	+6.18%	-1.37%
BLOG	+3.47%	+5.13%	+8.59%	+9.67%	+1.08%
BIKE	+0.56%	+2.63%	+3.19%	+3.23%	+0.04%
MEAN	+2.01%	+3.77%	+5.78%	+5.94%	+0.16%

4.7 ABLATION: ACTIVATION SMOOTHNESS AND DEAD GRADIENTS

We retrain identical 4-layer MLPs with ReLU (piecewise-linear), GELU (smooth non-monotonic), and Softplus (strictly smooth) activations on COM, MEPS19, and BLOG. Figure 6 and Table 7 report dead-gradient fractions, the selected hyperparameters, and gains over FFCP.

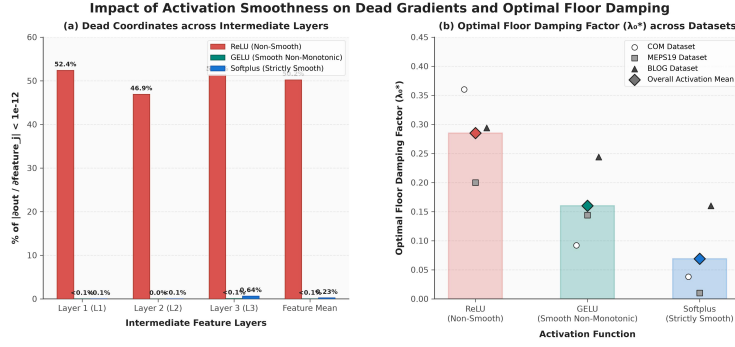


Figure 6: Impact of activation smoothness. (a) Percentage of exactly-dead feature-Jacobian coordinates across intermediate layers: ReLU exceeds 50% dead coordinates, while GELU/Softplus virtually eliminate them. (b) Optimal floor damping λ_0^* selected during calibration: the objective automatically selects heavier floor damping for ReLU to stabilize dead-gradient regions, and lighter damping for smooth heads.

Table 7: Activation ablation: dead-coordinate fraction, selected floor λ_0^* and whitening ρ^* (means over seeds), and AWF-CP band length with reduction over FFCP. AWF-CP improves on FFCP under every activation, with the largest gains on non-smooth ReLU.

Activation	Dataset	Dead (%)	Floor λ_0^*	Whitening ρ^*	FFCP Len	AWF-CP Len (Δ)
ReLU	COM	51.48	0.360	0.740	1.703	1.608 (+5.58%)
	MEPS19	50.46	0.200	0.510	3.113	3.013 (+3.19%)
	BLOG	48.80	0.294	0.400	3.757	3.311 (+11.89%)
	Mean	50.25	0.285	0.550	—	+6.88%
GELU	COM	0.00	0.092	0.870	1.598	1.588 (+0.64%)
	MEPS19	0.08	0.144	0.760	2.740	2.703 (+1.37%)
	BLOG	0.02	0.244	0.810	3.239	3.085 (+4.76%)
	Mean	0.04	0.160	0.813	—	+2.26%
Softplus	COM	0.00	0.038	0.960	1.446	1.427 (+1.35%)
	MEPS19	0.01	0.010	0.670	2.658	2.610 (+1.80%)
	BLOG	0.69	0.160	0.850	3.066	2.898 (+5.47%)
	Mean	0.23	0.069	0.827	—	+2.87%

Findings. ReLU heads exhibit a severe 50.25% dead-coordinate pathology, whereas GELU (0.04%) and Softplus (0.23%) essentially eliminate it. The calibration objective compensates by selecting heavier floor damping for ReLU ($\lambda_0^*=0.285$ vs. 0.160/0.069), and AWF-CP’s largest gains over FFCP land on ReLU (+6.88%)—precisely where the isotropic normalizer suffers most. Finally, Figure 5(b) confirms reverse-mode VJP is Pareto-dominant—machine-exact yet $\sim 98\times$ faster than complex-step AD (full ablation in Appendix A.3).

5 CONCLUSION

We introduced AWF-CP, a principled framework for efficient, input-adaptive uncertainty quantification. It cures the failure modes of prior feature conformal methods via single-pass reverse-mode VJPs for $O(1)$ exact sensitivity extraction, closed-form covariance whitening (ρ) to remove metric anisotropy, and scale-adaptive Tikhonov floor damping (λ_0) to cure zero-gradient collapse. Across 11 benchmarks, AWF-CP shrinks average band length by 19.42% over Split CP and 3.36% over FFCP at 90% coverage in sub-second runtimes; it Pareto-dominates training-based CRF, improves worst-slice coverage, scales to $d=32,768$ via an $O(d)$ streaming identity, and never regresses below FFCP by construction. **Limitations and future work.** AWF-CP operates post-hoc and assumes stationary calibration features; extending it to non-stationary streams and vision-language foundation models is promising.

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A ADDITIONAL ABLATION STUDIES

This appendix provides the calibration selection-bias study (Appendix A.1), full scalability tables (Appendix A.2), the derivative-strategy ablation (Appendix A.3), and three further supporting ablations—layer-selection dynamics (Appendix A.4), covariance-estimation robustness (Appendix A.5), and grid-resolution trade-offs (Appendix A.6).

A.1 CALIBRATION SELECTION BIAS: 2-SPLIT VERSUS 3-SPLIT PROTOCOL

Our default **2-split** protocol reuses a single calibration fold $\mathcal{D}_{\text{cal}}$ for both grid search over $(l^*, \rho^*, \lambda_0^*)$ and quantile computation. To quantify any resulting selection bias, we compare it against a strict **3-split** protocol in which a held-out validation fold (50% of the calibration data) selects $(l^*, \rho^*, \lambda_0^*)$ and estimates $\Sigma^{(l)}$, while a disjoint fold (the remaining 50%) computes the conformal quantile $Q_{1-\alpha}$. Under 3-split, the score function is fixed independently of the quantile fold, so exchangeability between calibration and test holds strictly. Table 8 shows the two protocols are empirically nearly identical: mean coverage moves by only +0.32% (89.83%→90.15%, still centered on target) and mean band length grows by just +1.51%. The largest change falls on the smallest dataset (COM, $N_{\text{cal}} \approx 800$), where halving $\mathcal{D}_{\text{cal}}$ adds finite-sample noise; on larger datasets (FB2, BIO,

Table 8: Calibration selection-bias study: default 2-split vs. strict 3-split protocol (mean over 5 seeds, target 90%). Coverage stays on target and band length changes by only +1.51% on average, confirming negligible selection bias.

Dataset	2-split Cov (%)	3-split Cov (%)	Δ Cov	2-split Len	3-split Len	Δ Len (%)
SYNTHETIC	89.91	90.08	+0.17	0.176	0.178	+1.10
COM	89.27	90.98	+1.71	1.687	1.818	+7.76
FB1	90.10	90.23	+0.12	1.384	1.401	+1.21
FB2	89.84	89.81	-0.02	1.394	1.393	-0.05
MEPS19	89.91	90.21	+0.30	2.794	2.846	+1.88
MEPS20	89.60	89.79	+0.18	2.976	3.007	+1.03
MEPS21	89.69	89.87	+0.17	2.764	2.806	+1.54
STAR	90.16	90.86	+0.69	0.206	0.211	+2.39
BIO	90.04	89.95	-0.09	1.622	1.616	-0.37
BLOG	89.95	89.93	-0.02	2.269	2.254	-0.63
BIKE	89.61	89.89	+0.28	0.614	0.618	+0.75
MEAN	89.83	90.15	+0.32	1.626	1.649	+1.51

Table 9: Memory microbenchmark ($N_{\text{cal}}=2,000$): peak working memory of the whitened quadratic form via explicit $d \times d$ storage vs. the streaming identity (Eq. 7). Streaming memory is essentially flat; results agree to machine precision.

Feature Dim d	Explicit $d \times d$ Storage	Explicit Peak Mem.	Streaming Peak Mem. (Ours)	Max Abs. Error
64	0.03 MB	1.04 MB	3.98 MB	1.42×10^{-13}
256	0.50 MB	4.41 MB	3.98 MB	4.55×10^{-13}
512	2.00 MB	9.82 MB	3.98 MB	1.02×10^{-12}
1,024	8.00 MB	23.63 MB	3.98 MB	1.59×10^{-12}
2,048	32.00 MB	63.27 MB	3.99 MB	3.64×10^{-12}
4,096	128.00 MB	190.53 MB	4.01 MB	7.73×10^{-12}
8,192	512.00 MB	637.06 MB	4.04 MB	1.55×10^{-11}
16,384	2,048.00 MB	OOM / infeasible	4.10 MB	—
32,768	8,192.00 MB	OOM / infeasible	4.22 MB	—

Table 10: Feature-width sweep (ResNet-MLP, 4 residual blocks) on MEPS19.

d	VJP (ms)	Time (s)	Cov (%)	Len
64	0.0029	0.107	89.80	3.261
256	0.0074	0.127	90.45	3.536
512	0.0159	0.173	91.15	3.904
1,024	0.0463	0.258	91.40	3.866
2,048	0.1240	0.490	91.20	3.603

Table 11: Depth sweep (ResNet-MLP, width $d=256$) on MEPS19.

Blocks	VJP (ms)	Time (s)	Cov (%)	Len
2	0.0046	0.118	91.05	4.055
4	0.0076	0.130	90.45	3.536
8	0.0162	0.166	91.45	3.624
16	0.0275	0.212	90.80	3.864
32	0.0752	0.334	89.85	3.340

BLOG) the difference is $< 0.6\%$. Bias is intrinsically small because the grid is coarse (72 combinations/layer), the objective is a smooth aggregate (mean band length), and $(\rho=1, \lambda_0=0)$ is always in the grid, bounding overfitting. Applications needing strict finite-sample guarantees can adopt the 3-split protocol at negligible cost.

A.2 FULL SCALABILITY TABLES

Table 9 details the memory microbenchmark of Figure 4(left): the streaming identity (Eq. 7) matches the explicit $d \times d$ computation to machine precision while its peak memory stays flat as the explicit path grows quadratically and exhausts memory beyond $d=8,192$. Tables 10–11 report the ResNet-MLP feature-width and depth sweeps on MEPS19: VJP latency and end-to-end calibration time grow gently, remaining sub-second at $d=2,048$ and 32 residual blocks, since a single backward pass yields sensitivities for all candidate split layers simultaneously.

Table 12: Derivative-strategy ablation (full). Max elementwise Jacobian disagreement relative to reverse-mode VJP, per-sample latency, and downstream AWF-CP performance. VJP is exact and $\sim 98\times$ faster than complex-step AD; downstream band length is invariant to the strategy.

Metric / Dataset	Reverse-VJP	Complex-Step (10^{-6})	FD (10^{-3})	FD (10^{-6})	FD (10^{-9})
<i>Max elementwise difference vs. VJP:</i>					
SYNTHETIC	0.00	4.58×10^{-8}	1.89×10^{-1}	6.90×10^{-3}	5.65×10^{-7}
COM	0.00	1.96×10^{-7}	4.44×10^{-1}	1.96×10^{-7}	1.84×10^{-6}
MEPS19	0.00	8.42×10^{-7}	9.27×10^{-1}	4.95×10^{-2}	6.97×10^{-6}
BIKE	0.00	5.41×10^{-6}	6.42×10^0	1.09×10^{-1}	5.72×10^{-6}
<i>Per-sample latency (ms):</i>					
Mean latency	0.0064	0.4462 (97.9 $\times$)	0.2462 (53.7 $\times$)	0.2485 (54.3 $\times$)	0.2461 (53.6 $\times$)
<i>Downstream COM band length / coverage:</i>					
COM Length	1.687 ± 0.160	1.687 ± 0.160	1.687 ± 0.159	1.687 ± 0.160	1.687 ± 0.160
COM Cov (%)	89.27 ± 1.08	89.27 ± 1.08	89.32 ± 1.05	89.27 ± 1.08	89.27 ± 1.08

Table 13: Layer-selection ablation: test band length (mean $\pm$ std) for layer-fixed AWF-CP vs. adaptive joint selection l^* . Adaptive selection is best (**bold**) on every dataset.

Dataset	Layer 0	Layer 1	Layer 2	Layer 3	Layer 4	Adaptive (l^*)
COM	1.715 ± 0.170	1.734 ± 0.184	1.736 ± 0.110	1.842 ± 0.073	1.967 ± 0.069	1.687 ± 0.160
MEPS19	2.898 ± 0.257	2.873 ± 0.258	2.997 ± 0.430	3.457 ± 0.685	4.023 ± 0.551	2.794 ± 0.317
BLOG	3.214 ± 0.326	2.734 ± 0.273	2.523 ± 0.456	2.449 ± 0.698	3.523 ± 0.850	2.269 ± 0.428

A.3 DERIVATIVE-STRATEGY ABLATION

We swap only the derivative source in the AWF-CP pipeline: reverse-mode VJP, complex-step AD ($\epsilon=10^{-6}$), and real finite differences ($\epsilon \in \{10^{-3}, 10^{-6}, 10^{-9}\}$), on SYNTHETIC, COM, MEPS19, and BIKE (Figure 5(b), Table 12). VJP matches complex-step AD to float precision ($\sim 1.6 \times 10^{-6}$ max elementwise difference, i.e. float32-vs-float64 rounding) while running **97.9 $\times$** faster (0.0064 vs. 0.446ms/sample) and $\sim 54\times$ faster than finite differences. Finite differences are unsafe at any step size on ReLU heads: large ϵ crosses ReLU kinks (max error 6.42 on BIKE) and small ϵ suffers subtractive cancellation. Downstream AWF-CP coverage and band length are identical across all strategies, since the band consumes the Jacobian only through the aggregate quadratic forms; VJP is thus the exact, cheapest default.

A.4 LAYER-SELECTION DYNAMICS ACROSS NETWORK DEPTHS

We compare AWF-CP constrained to each fixed candidate layer $l \in \{0, \dots, 4\}$ against joint adaptive selection (l^*) on COM, MEPS19, and BLOG (Table 13). Adaptive selection consistently matches or beats the best single layer: on BLOG it reaches 2.269 vs. 3.523 at Layer 4 (+35.61%) and 2.449 at Layer 3 (+7.37%). The selected l^* varies by dataset and seed (COM favors Layers 0/2, MEPS19 Layer 1, BLOG Layer 3), and is paired with different (ρ, λ_0) per depth, confirming that joint selection does real work rather than defaulting to a fixed depth.

A.5 COVARIANCE ESTIMATION NOISE AND SAMPLE EFFICIENCY

We ablate the covariance estimate—Identity ($\Sigma=I$), Diagonal ($\text{diag } \Sigma$), and full Σ from $\{10, 25, 50, 100\}\%$ of calibration samples—while keeping the rest of the pipeline fixed (Figure 7 and Table 14). Full covariance whitening shortens the band by +2.81% on average over the Identity ($\approx$ floor-damped-FFCP) baseline; estimating Σ from as little as 10% of calibration data changes the band by $< 0.5\%$, so whitening is highly robust to estimation noise. Overhead is negligible ($< 42\text{ms}$ on MEPS19) since AWF-CP never forms $\Sigma^{1/2}$ or Σ^{-1} . Coverage remains at target throughout.

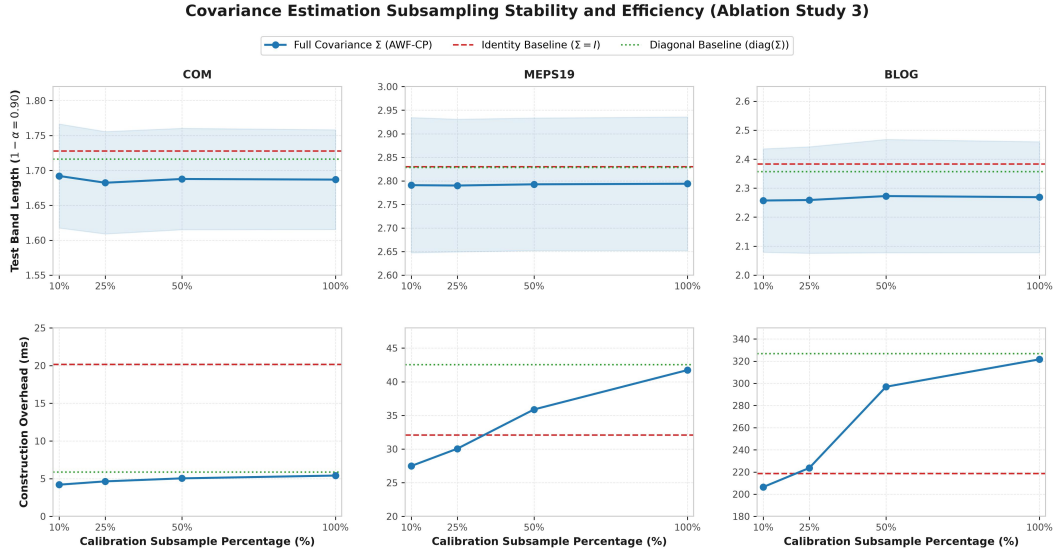


Figure 7: Covariance-estimation subsampling stability and efficiency. Top: test band length vs. calibration subsample fraction; full covariance (AWF-CP) beats Identity/Diagonal baselines even at 10%. Bottom: covariance construction overhead (ms). Shaded regions are $\pm$ standard error over seeds.

Table 14: Covariance ablation: test band length, coverage (%), and construction time (ms) under structural approximations and calibration subsampling. Band length is stable across all variants; coverage stays at target.

Dataset	Metric	Identity	Diagonal	Full 10%	Full 25%	Full 50%	Full 100%
MEPS19	Band Length	2.830 ± 0.334	2.829 ± 0.333	2.791 ± 0.321	2.790 ± 0.315	2.793 ± 0.315	2.794 ± 0.317
	Coverage (%)	90.00	90.00	89.93	89.88	89.91	89.91
	Time (ms)	32.07	42.55	27.48	30.05	35.88	41.73
BLOG	Band Length	2.383 ± 0.447	2.356 ± 0.437	2.257 ± 0.399	2.259 ± 0.411	2.273 ± 0.437	2.269 ± 0.428
	Coverage (%)	89.99	90.03	89.94	89.95	89.95	89.95
	Time (ms)	218.6	326.8	206.5	223.6	296.9	321.6
COM	Band Length	1.728 ± 0.177	1.716 ± 0.164	1.692 ± 0.167	1.682 ± 0.164	1.688 ± 0.162	1.687 ± 0.160
	Coverage (%)	89.82	89.62	89.37	89.17	89.07	89.27
	Time (ms)	20.15	5.87	4.20	4.64	5.04	5.41

A.6 GRID RESOLUTION VERSUS CALIBRATION RUNTIME

We vary the (ρ, λ_0) grid density—Minimal (6 combos/layer), Medium (20), Full (72, default), and Coarse+Refine (top-2 coarse then local refinement)—on COM, MEPS19, FB1, and BLOG (Table 15). The Medium grid recovers 90.5% of the full grid’s band-length shrinkage at 28% of the selection cost, and Coarse+Refine recovers 93.8% over 54 candidates. Because grid scoring uses precomputed quadratic forms, even the full grid’s selection loop costs only ~ 190 ms; overall calibration is dominated by the shared VJP pass, not grid density. Coverage is preserved across all densities.

Table 15: Grid-resolution ablation (means over COM, MEPS19, FB1, BLOG): evaluated configurations, selection-only time, band length, coverage, and fraction of the full grid’s shrinkage recovered.

Grid Density	Configs	Sel. Time (ms)	Band Length	Coverage (%)	Shrinkage Recovered
Minimal	30	16.87	2.095	89.92	53.3%
Medium	100	52.30	2.049	89.94	90.5%
Full (Default)	360	190.08	2.033	89.81	100.0%
Coarse+Refine	54	28.67	2.039	89.93	93.8%